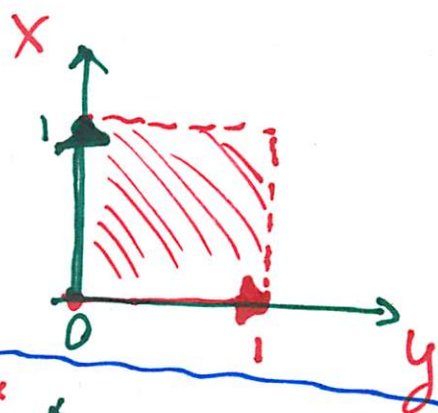


EC 400 - 2013 - LECTURE VISUALIZER

IDENTITY:

$$\begin{matrix} x & A_1 & A_2 \\ y & \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{matrix} = A$$

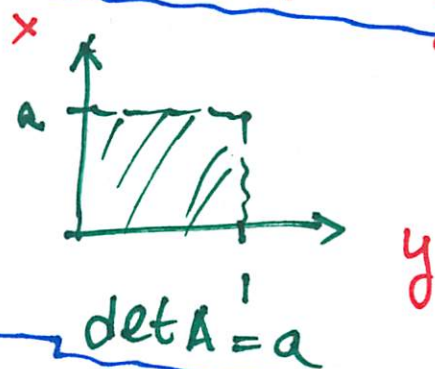
• $\det I = 1$



LINEARITY:

$$A = \begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix}$$

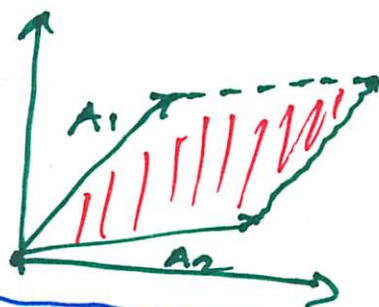
$$\det(aA_1, A_2) = \det(A_1, A_2) \cdot a$$



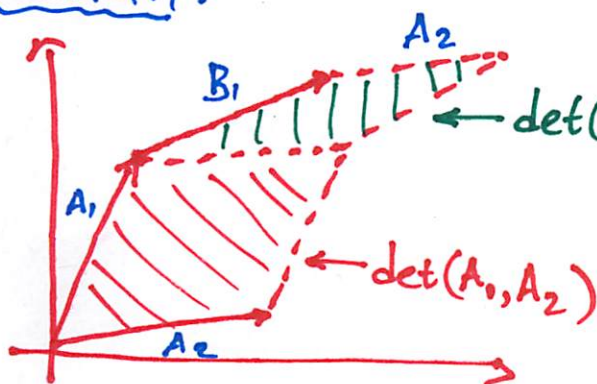
GENERAL CASE:

$$A = [A_1, A_2]$$

RED AREA IS
DET



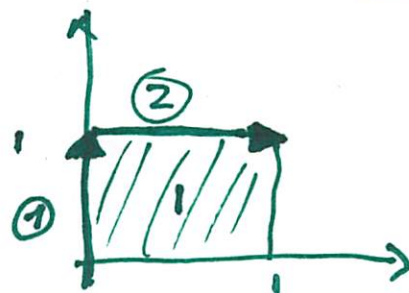
LINEARITY:



$$\begin{aligned} \det(A_1 + B_1, A_2) &= \det(A_1, A_2) \\ &+ \det(B_1, A_2) \end{aligned}$$

CHANGING ORDER:

$$\det(A_1, A_2) = -\det(A_2, A_1)$$



$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

SIGNING DIAGONAL QUADRATIC FORMS:

$$Q(x) = a_{11} x_1^2 + a_{22} x_2^2 + a_{33} x_3^2$$

<u>PD</u> $\Rightarrow a_{11}, a_{22}, a_{33} > 0$	<u>ND</u> $a_{11}, a_{22}, a_{33} < 0$
1 ST LPM $a_{11} > 0$	$a_{11} < 0$
2 ND LPM $\begin{vmatrix} a_{11} & 0 \\ 0 & a_{22} \end{vmatrix} = a_{11} a_{22} > 0 \Rightarrow a_{22} > 0$	$a_{11} a_{22} > 0 \Rightarrow a_{22} < 0$
3 RD LPM $\begin{vmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{vmatrix} = a_{11} a_{22} a_{33} > 0 \Rightarrow a_{33} > 0$	$a_{11} a_{22} a_{33} < 0 \Rightarrow a_{33} < 0$

FOR SEMIDEF INEQUALITIES SUFFICES...

DEFINITENESS & 2X2 MATRICES:

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

$$Q(x) = a x_1^2 + 2b x_1 x_2 + c x_2^2$$

ADD & SUBTRACT
 $\frac{b^2 x_2^2}{a}$

$$= \left[a x_1^2 + 2b x_1 x_2 + \frac{b^2 x_2^2}{a} \right] + \left(c - \frac{b^2}{a} \right) x_2^2$$

$$= a \left(x_1 + \frac{b}{a} x_2 \right)^2 + \left(\frac{ac - b^2}{a} \right) x_2^2$$

NEED:

• $a \neq 0$ TO DO THIS!

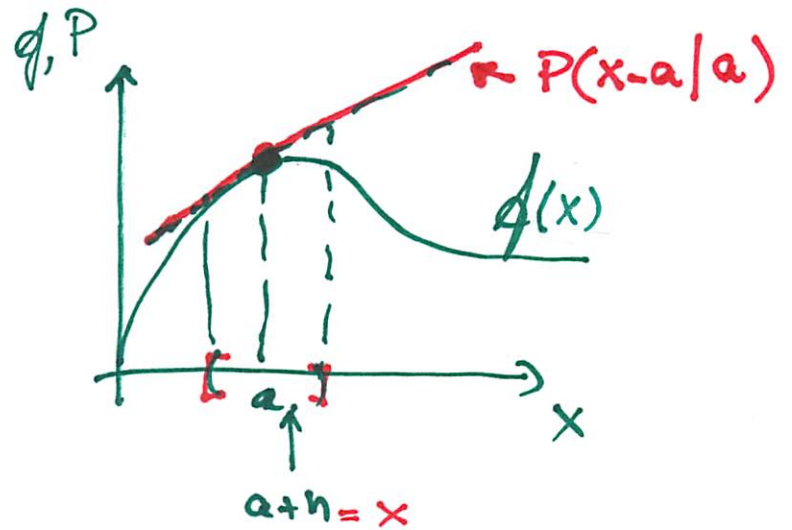
• POSIT $ac - b^2 \neq 0$

$\Rightarrow$ CASE 1: $a > 0$ & $ac > b^2 \Rightarrow Q(x) \geq 0$ $\Rightarrow$ PD $\uparrow$ ND
MOREOVER: $Q(x) = 0 \Leftrightarrow x_2 = x_1 = 0$
CASE 2: $a < 0$ & $ac > b^2 \Rightarrow Q(x) \leq 0$ & $Q(x) = 0 \Leftrightarrow x = 0$

$$\lim_{h \rightarrow 0} \frac{R(h|a)}{h} = \lim_{h \rightarrow 0} \underbrace{\left[\frac{\phi(a+h) - \phi(a)}{h} \right]}_{L = \phi'(a)} - \phi'(a)$$

FIRST ORDER
TAYLOR EXPANSION

BEST LOCAL APPROX OF
DEGREE 1



EQUIVALENT DEFS OF CONCAVITY:

• CONCAVE $\Leftrightarrow f(y) - f(x) \leq f'(x)(y-x)$ $f \text{ is } D'$ *

PF $\Rightarrow$:

$$f(ty + (1-t)x) \geq tf(y) + (1-t)f(x)$$

$$f(x + t(y-x)) \geq f(x) + t(f(y) - f(x))$$

$$\frac{f(x + t(y-x)) - f(x)}{t(y-x)} \geq \frac{f(x+h) - f(x)}{h} \quad (y-x) \geq f(y) - f(x)$$

→ TAKE $\lim_{h \rightarrow 0} \Rightarrow f'(x)(y-x) \geq f(y) - f(x)$

$\Leftarrow$: LET: $z = ty + (1-t)x \quad \forall t \in [0,1]$ □

$$\begin{aligned} (*) \Rightarrow & \left[\begin{aligned} f(y) - f(z) &\leq f'(z)(y-z) \\ f(x) - f(z) &\leq f'(z)(x-z) \end{aligned} \right] \begin{matrix} t \\ (1-t) \end{matrix} \quad (+) \end{aligned}$$

$$\Rightarrow tf(y) + (1-t)f(x) - f(z) \leq f'(z) [\cancel{ty + (1-t)x} - z] = 0$$

$$\Rightarrow tf(y) + (1-t)f(x) \leq f(z) \quad \blacksquare$$

• CONCAVE $\iff \phi''(x) \leq 0 \quad \forall x$ IF ϕ IS D^2

PF $\implies$:

$$\left. \begin{aligned} \phi(y) - \phi(x) &\leq \phi'(x)(y-x) \\ \phi(x) - \phi(y) &\leq \phi'(y)(x-y) \end{aligned} \right] \textcircled{+}$$

$$0 \leq [\phi'(y) - \phi'(x)](x-y)$$

DEVIDE BY $-(x-y)^2$

$$0 \geq \frac{\phi'(y) - \phi'(x)}{y-x}$$

TAKIN' LIMS
AS $y \rightarrow x$

$\downarrow y \rightarrow x$

$$0 \geq \phi''(x)$$



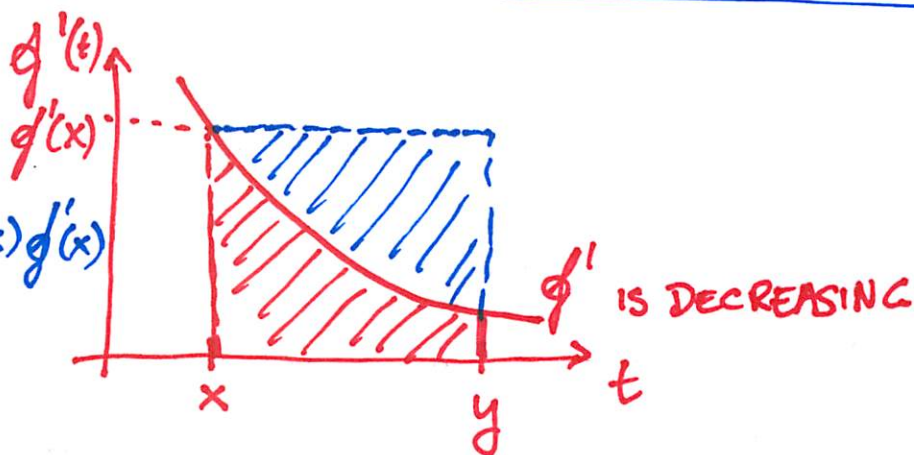
PF $\Leftarrow$: $\phi(y) - \phi(x) \stackrel{\text{FTC}}{=} \int_x^y \phi'(t) dt \leq$ AS $\phi'' \leq 0$

FOR $x < y$

$$\int_x^y \phi'(x) dt = \phi'(x)(y-x)$$

IN BLUE + RED:

$$\begin{aligned} \int_x^y \phi'(x) dt \\ = \int_x^y dt \phi'(x) = (y-x)\phi'(x) \end{aligned}$$



IN RED

$$\int_x^y \phi'(t) dt$$

• Q-CONCAVE $\Leftrightarrow \phi(tx + (1-t)y) \geq \min\{\phi(x), \phi(y)\} \quad \forall x, y$
⊛

PF $\Rightarrow$ TAKE $\forall x, y \in U$

DEFINE $K = \min\{\phi(x), \phi(y)\}$

$\Rightarrow x, y \in C_K^+$

$\Rightarrow$ BY Q-CONCAVITY $tx + (1-t)y \in C_K^+$

$\Rightarrow \phi(tx + (1-t)y) \geq K = \min\{\phi(x), \phi(y)\}$ □

PF $\Leftarrow$ TAKE $\forall x, y \in C_K^+$

$\Rightarrow \min\{\phi(x), \phi(y)\} \geq K$

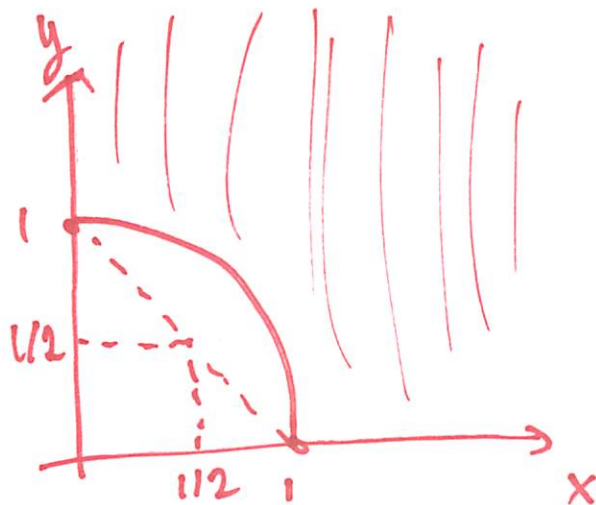
$\Rightarrow \phi(tx + (1-t)y)$ BY ⊛ FOR $\forall t \in [0, 1]$

$\Rightarrow tx + (1-t)y \in C_K^+ \quad \forall t \in [0, 1]$

$\Rightarrow C_K^+$ IS CONVEX □

• MONOTONE FUNCTION THAT IS NOT Q-CONCAVE

$$f(x, y) = x^2 + y^2 \text{ ON } \mathbb{R}_+^2$$



REMARK: INCREASE TRANSFORM OF Q-CONC FUNCTION IS Q-CONC

PF $C_k^+(h) = \{x \in U \mid h(x) \geq k\} \quad \forall h$

LET ϕ BE Q-CONC

LET $g: \mathbb{R} \rightarrow \mathbb{R}$ ST $g' > 0$

WANT

$\Rightarrow h(x) = g(\phi(x))$ IS Q-CONC

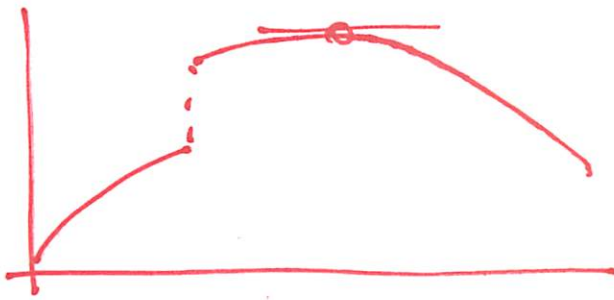
$$C_k^+(\phi) = C_{g(k)}^+(h)$$

$$\{x \in U \mid \phi(x) \geq k\} \quad \{x \in U \mid h(x) \geq g(k)\}$$

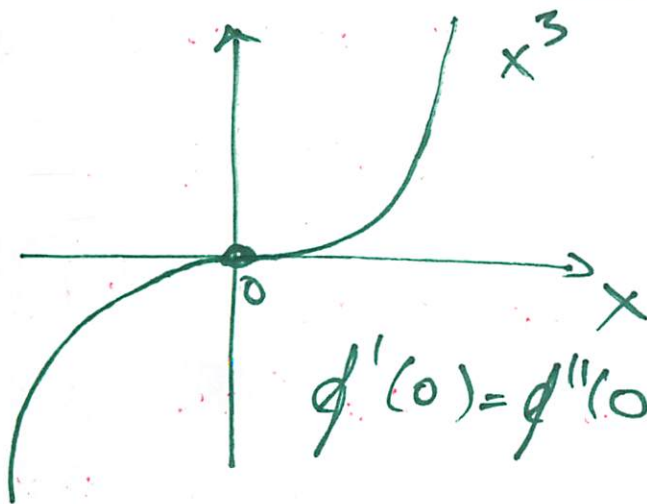
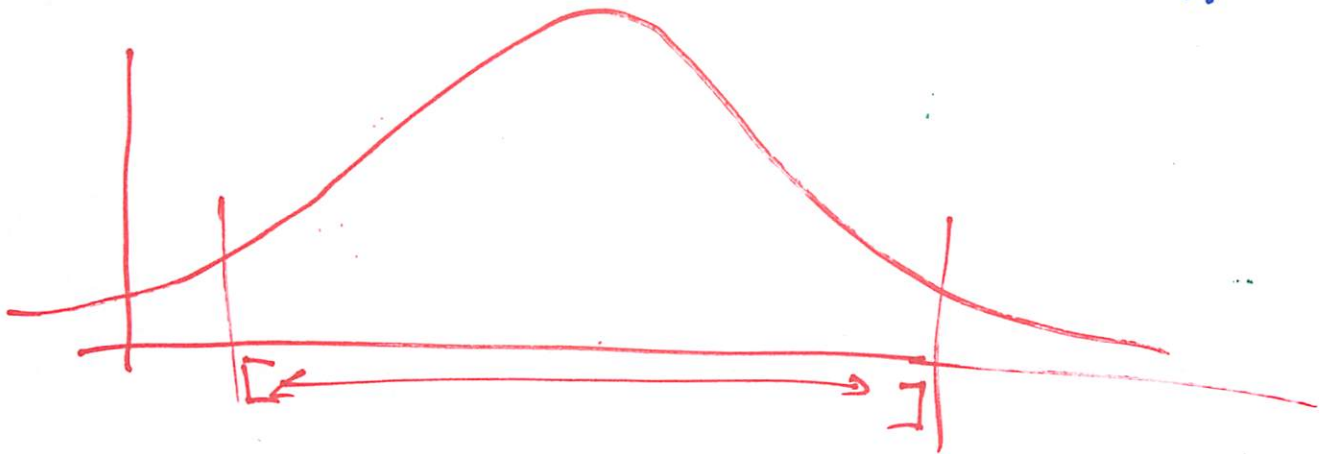
$\updownarrow$

SO AS $C_k^+(\phi)$ IS CONVEX, $\boxed{\phi(x) \geq k}$
 SO IS $C_{g(k)}^+(h) \quad \forall k$

OPTIMIZATION



WHEN DISCONTINUITY
OR UNBOUNDEDNESS
OF SUPPORT ARE NOT
A PROBLEM FOR
EXTREME VALUE
THM.



$$f'(0) = f''(0) = 0$$

← 1-DIMENSIONAL FUNCTION
FOR WHICH CRITICAL
POINT 0 IS NOT
EXTREME POINT
DESPITE $f''(0) = 0$

REMARK: IF $D\phi(x^*) = 0$ & $D^2\phi(x^*) \underset{\text{IS ND}}{< 0}$

$\Rightarrow x^*$ IS STRICT LOCAL MAX.

PF: TAYLOR EXPAND 2ND ORDER:

$$\phi(x^* + h) = \phi(x^*) + \underbrace{D\phi(x^*)}_{L=0} h + \frac{1}{2} h^T D^2\phi(x^*) h$$

$$\Rightarrow \phi(x^* + h) \approx \phi(x^*) + \underbrace{\frac{1}{2} h^T D^2\phi(x^*) h}_{L < 0 \text{ AS } D^2\phi(x^*) < 0} + \underbrace{R_2(h|x^*)}_{\substack{\text{NEGLECT} \\ \text{AS } h \text{ IS} \\ \text{SMALL ENOUGH}}}$$

$$\Rightarrow \phi(x^* + h) - \phi(x^*) < 0 \text{ FOR } h \text{ SMALL} \quad \square$$

GLOBAL SOC PROOF

• IF ϕ IS D', CONCAVE $\Leftrightarrow D\phi(x)(y-x) \geq \phi(y) - \phi(x)$

$$\Rightarrow 0 \geq D\phi(x)(y-x) \geq \phi(y) - \phi(x)$$

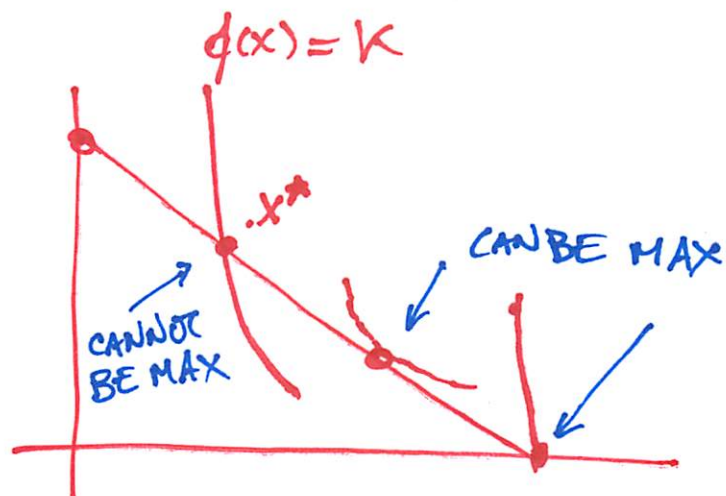
$$\Rightarrow \phi(x) \geq \phi(y) \quad \square$$

• BUT THEN IF $D\phi(x^*) = 0$

$$\Rightarrow 0 = D\phi(x^*)(y-x^*) \geq \phi(y) - \phi(x^*) \quad \square$$

$\forall y$ $\forall y$

CONSTRAINED OPTIMIZATION



SLOPE LEVEL CURVE:

$$\Rightarrow f(x_1, x_2) = a$$

$$f(x_1, x_2(x_1)) = a$$

SOLVE $\frac{\partial x_2}{\partial x_1}$ BY DIFF WRT x_1

TOT DIFF

$$\frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 = 0$$

$$\Rightarrow dx_2 \frac{\partial f}{\partial x_2} = - \frac{\partial f}{\partial x_1} dx_1$$

$$\Rightarrow \frac{dx_2}{dx_1} = - \frac{\partial f / \partial x_1}{\partial f / \partial x_2}$$

SLOPE CONSTRAINT:

FOR CONSTRAINT $h(x_1, x_2) = 0$

$$\Rightarrow \frac{dx_2}{dx_1} = - \frac{\partial h / \partial x_1}{\partial h / \partial x_2}$$

TANGENCY:

$$\Rightarrow \frac{\partial h / \partial x_1}{\partial h / \partial x_2} = \frac{\partial f / \partial x_1}{\partial f / \partial x_2} \Rightarrow \frac{\partial f / \partial x_1}{\partial h / \partial x_1} = \frac{\partial f / \partial x_2}{\partial h / \partial x_2} = \mu$$

$$\mathcal{L}(x) = f(x) - \mu h(x)$$

GRADIENT OF $h(x)$:
 $m \times 1$

$$Dh(x) = \begin{bmatrix} \partial h_1 / \partial x_1 & \dots & \partial h_1 / \partial x_n \\ \vdots & & \\ \partial h_m / \partial x_1 & \dots & \partial h_m / \partial x_n \end{bmatrix}$$

$m \times n$

PROOF LAGRANGE THM:

$$\textcircled{1} \quad \phi^a(x) = \phi(x) - \frac{a}{2} \frac{\|h(x)\|^2}{h(x)^T h(x)} - \frac{c}{2} \frac{\|x - x^*\|^2}{(x - x^*)^T (x - x^*)}$$

LET x^a BE THE MAXIMIZER OF $\phi^a(x)$

FOR ALL x ST $\|x - x^*\| \leq \epsilon$

$$\textcircled{2} \quad \text{DEFINE } \bar{x} = \lim_{a \rightarrow \infty} x^a \Rightarrow \|h(\bar{x})\| = 0$$

AS ϕ IS BOUNDED IN THE BALL AROUND x^*

$$\phi^a(\bar{x}) = \phi(\bar{x}) - \frac{c}{2} \|\bar{x} - x^*\|^2 = \lim_{a \rightarrow \infty} \phi^a(x^a) \leq \phi(x^*)$$

$\Rightarrow$ IT MUST BE $\bar{x} = x^*$, AS $\|x^* - x^*\| = 0 \leq \epsilon$.

$\textcircled{3}$ FOC FOR ϕ^a REQUIRE & MUST HOLD @ x^a :

$$D\phi^a(x^a) = \underbrace{D\phi(x^a)}_{1 \times n} - a \underbrace{h(x^a)^T}_{1 \times m} \underbrace{Dh(x^a)}_{m \times n} - c \underbrace{(x^a - x^*)^T}_{1 \times n} = 0_{1 \times n}$$

$\textcircled{4}$ SOLVE FOC:

$$a h(x^a)^T Dh(x^a) = [D\phi(x^a) - c(x^a - x^*)^T]$$

POST MULT: $Dh^T (Dh Dh^T)^{-1}$

$$\Rightarrow a h(x^a)^T = [D\phi(x^a) - c(x^a - x^*)^T] Dh(x^a)^T (Dh(x^a) Dh(x^a)^T)^{-1}$$

TAKING
LIMITS

$\downarrow$
 x^*

$$= D\phi(x^*) Dh(x^*)^T (Dh(x^*) Dh(x^*)^T)^{-1}$$

CONSTRAINED OPTIMIZATION EXAMPLE

$$\text{maximize } 2\sqrt{x} - x \quad \text{st } x \leq a$$

$$\text{KKT CONDITIONS } \mathcal{L} = 2\sqrt{x} - x - \mu(x - a)$$

$$\bullet \frac{1}{\sqrt{x}} - 1 - \mu = 0 \quad \text{FOC}$$

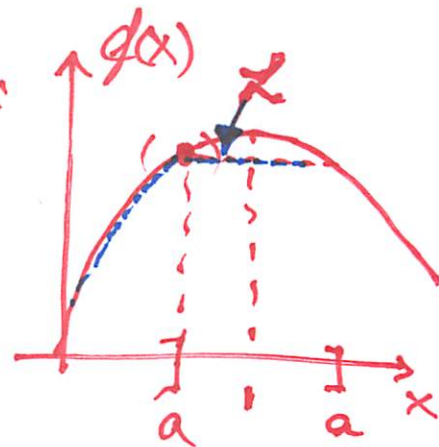
$$\bullet \mu(x - a) = 0 \quad \text{CS}$$

$$\bullet x - a \leq 0 \quad \mu \geq 0 \quad \text{NN}$$

• FOR $\mu = 0$ CONSTRAINT IS SLACK:

$$\Rightarrow \text{FOC} \Rightarrow x^* = 1$$

$$\text{NN} \Rightarrow x^* = 1 \leq a$$



• FOR $\mu > 0$ CONSTRAINT BINDS:

$$\Rightarrow \text{CS} \Rightarrow x^* = a$$

$$\Rightarrow \text{FOC} \Rightarrow \mu^* = \phi'(x^*) = \frac{1}{\sqrt{x^*}} - 1 = \frac{1}{\sqrt{a}} - 1$$

$$\Rightarrow \mu^* \geq 0 \Rightarrow \frac{1}{\sqrt{a}} \geq 1 \Rightarrow a \leq 1$$

$$\mathcal{L} = \phi(x) - \mu(x - a)$$

PROOF KKT SUFFICIENCY

① BY CS $\Rightarrow \phi(x^*) = \phi(x^*) - \overbrace{\mu^* g(x^*)}^{\Gamma=0}$

② AS x^* MAXES LAGRANGIAN $\Rightarrow \phi(x) - \underbrace{\mu^* g(x)}_{\Gamma \leq 0}$

③ AS $\mu^* \geq 0$
FOR $\forall x$ ST $\Rightarrow \geq \phi(x)$

$g(x) \leq 0$

$\Rightarrow \mu^* g(x) \leq 0$

$\Rightarrow$ SO x^* MAXES ϕ ST $g(x) \leq 0$

PROOF OF ENVELOPE THM:

$$\frac{dv(b)}{db} = \frac{d\phi(x(b)|b)}{db} =$$

$$= \underbrace{D_x \phi(x(b)|b)}_{1 \times n} \underbrace{D_b x(b)}_{n \times 1} + D_b \phi(x(b)|b)$$

$$= \overbrace{\mu(b) D_x g(x(b)|b)}^{|| \text{ BY FOC } ||} D_b x(b) + D_b \phi(x(b)|b)$$

$1 \times m \quad m \times n$

$$= \mu(b) \overbrace{(-D_b g(x(b)|b))}^{|| \text{ BY } (*) ||} + D_b \phi(x(b)|b)$$

$$= D_b \lambda(x(b), \mu(b)|b)$$

• AS CONSTRAINTS BIND $\downarrow (1) \quad \downarrow (2)$
TOTALLY DIFF WRT $b \Rightarrow g(x(b)|b) = 0$

$$\underbrace{D_x g(x(b)|b)}_{m \times n} \underbrace{D_b x(b)}_{n \times 1} + D_b g(x(b)|b) = 0 \quad (*)$$

$$\lambda(x(b), \mu(b)|b) = \underbrace{\phi(x(b)|b)}_{(1)} - \mu(b) \underbrace{g(x(b)|b)}_{(2)}$$

ENVELOPE THM FOR UNCONSTRAINED PROBLEMS:

$$\max_x \phi(x|b)$$

ENVELOPE: $\frac{d v(b)}{d b} = \frac{d \phi(x(b)|b)}{d b} = \frac{\partial \phi(x(b)|b)}{\partial b}$

PF: $\frac{d \phi(x(b)|b)}{d b} = D_x \phi(x(b)|b) D_b x(b) + D_b \phi(x(b)|b)$

$= 0$
BY FOC

I'M INTERESTED AT $\frac{d \phi(b)}{d b_j}$?

BY IFT

$\Rightarrow$

$$F(\phi(b)|b) = 0$$

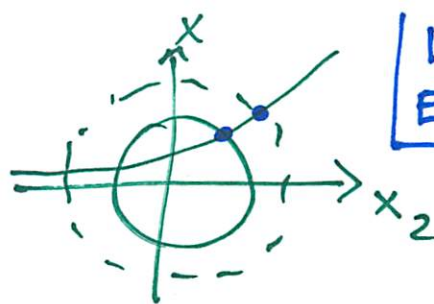
IMPLICIT
FUNCTION
THEOREM
SLOPES

TOT DIFF:
WRT b_j

$$D_x F(\phi(b)|b) D_{b_j} \phi(b) + D_{b_j} F(\phi(b)|b) = 0$$

$$\Rightarrow D_{b_j} \phi(b) = - D_x F(\phi(b)|b)^{-1} D_{b_j} F(\phi(b)|b) \quad \square$$

$$F(x|b) = \begin{bmatrix} x_1 - e^{x_2} \\ x_1^2 + x_2^2 - b \end{bmatrix}$$



IFT
EXAMPLE

$$\Rightarrow D_x F(x|b) = \begin{bmatrix} 1 & -e^{x_2} \\ 2x_1 & 2x_2 \end{bmatrix} \Rightarrow \det = 2x_2 + 2x_1 e^{x_2} \neq 0 \text{ FOR } x > 0$$

SO BY IFT $\exists \phi_1(b), \phi_2(b)$ ST

$$\begin{cases} \phi_1(b) - e^{\phi_2(b)} = 0 \\ \phi_1(b)^2 + \phi_2(b)^2 - b = 0 \end{cases}$$

• DIFF WRT b THE SYSTEM :

$$\begin{cases} \phi_1'(b) - \phi_2'(b) e^{\phi_2(b)} = 0 \\ 2\phi_1'(b)\phi_1(b) + 2\phi_2'(b)\phi_2(b) - 1 = 0 \end{cases}$$

Solving:

$$\Rightarrow 2\phi_2'(b) [\phi_1(b) e^{\phi_2(b)} + \phi_2(b)] = 1$$

$$\Rightarrow \phi_2'(b) = \frac{1}{2e^{\phi_2(b)}\phi_1(b) + 2\phi_2(b)} > 0$$

$$\Rightarrow \phi_1'(b) = e^{\phi_2(b)} \phi_2'(b) > 0$$